

1. Dim_Date - Applied Steps panel

The screenshot shows the Power BI Desktop interface with the 'Smart City Bike-Sharing - Station Performance Dashboard' open. The 'Fact_StationStatus' table is selected in the 'Queries' pane. The 'Applied Steps' panel on the right shows the 'Changed Type' step. The main view displays a table with columns: DateKey, Date, Year, Quarter, and Month. The table contains 21 rows of data, showing dates from 20220316 to 20220404.

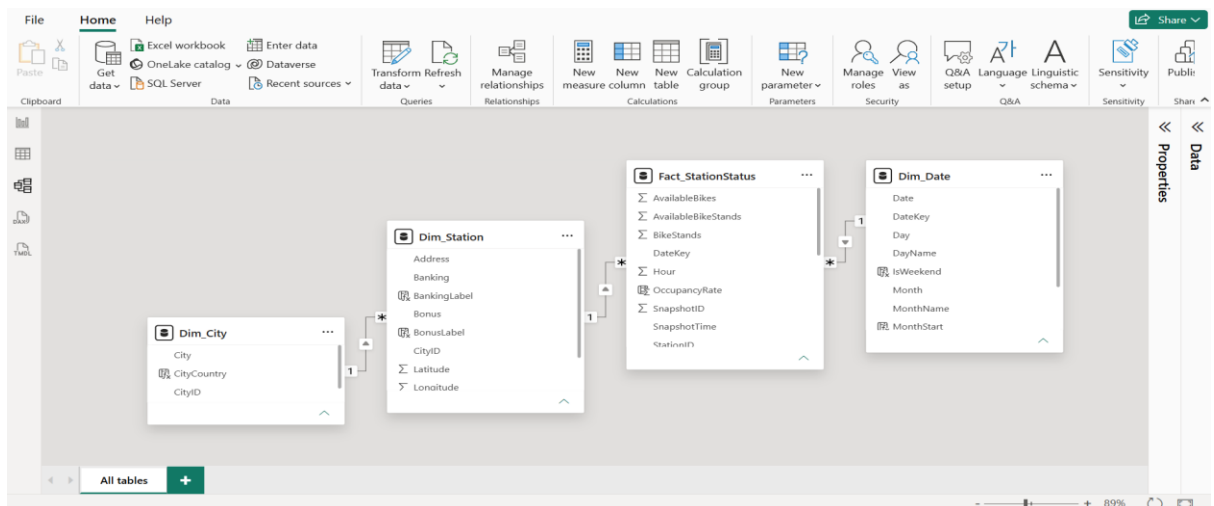
DateKey	Date	Year	Quarter	Month
20220316	16-03-2022	2022	1	1
20220317	17-03-2022	2022	1	1
20220318	18-03-2022	2022	1	1
20220319	19-03-2022	2022	1	1
20220320	20-03-2022	2022	1	1
20220321	21-03-2022	2022	1	1
20220322	22-03-2022	2022	1	1
20220323	23-03-2022	2022	1	1
20220324	24-03-2022	2022	1	1
20220325	25-03-2022	2022	1	1
20220326	26-03-2022	2022	1	1
20220327	27-03-2022	2022	1	1
20220328	28-03-2022	2022	1	1
20220329	29-03-2022	2022	1	1
20220330	30-03-2022	2022	1	1
20220331	31-03-2022	2022	1	1
20220401	01-04-2022	2022	2	2
20220402	02-04-2022	2022	2	2
20220403	03-04-2022	2022	2	2
20220404	04-04-2022	2022	2	2

2. Fact_StationStatus - Applied Steps panel

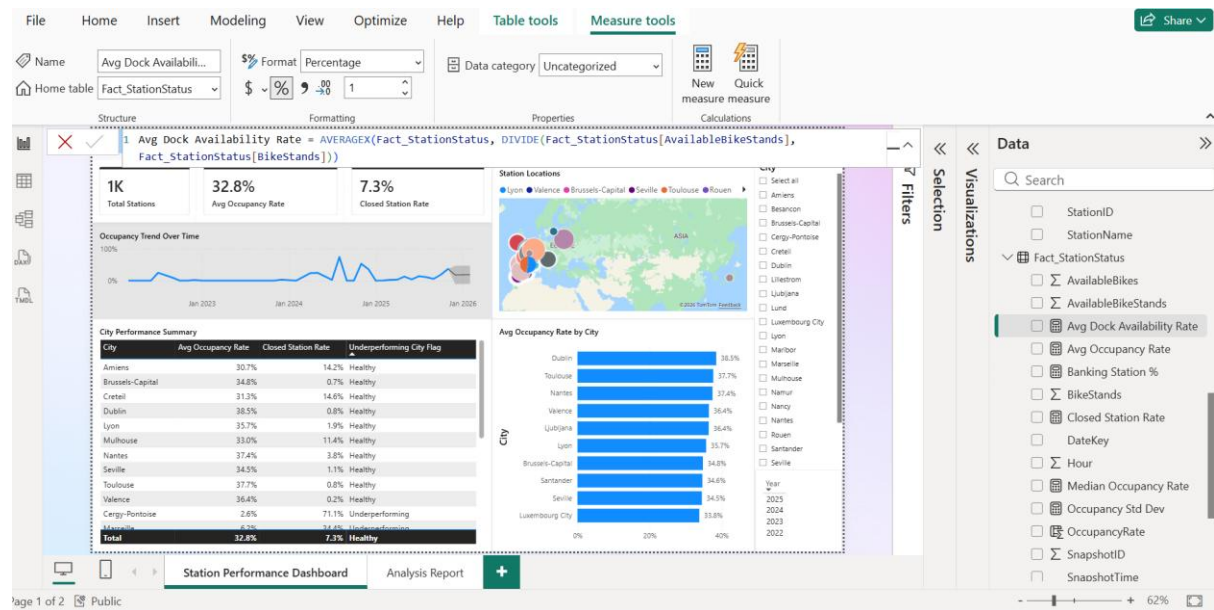
The screenshot shows the Power BI Desktop interface with the 'Fact_StationStatus' table selected in the 'Queries' pane. The 'Applied Steps' panel on the right shows the 'Changed Type' step. The main view displays a table with columns: SnapshotID, StationID, DateKey, SnapshotTime, and Hour. The table contains 21 rows of data, showing snapshots from 20220406 to 20221122.

SnapshotID	StationID	DateKey	SnapshotTime	Hour
1297	2	20220406	06-04-2022 12:54:35	1
8265	2	20220927	27-09-2022 10:57:48	2
8065	3	20220608	08-06-2022 11:50:13	3
4342	4	20220913	13-09-2022 11:28:01	4
6176	5	20220906	06-09-2022 18:02:17	5
3322	6	20220914	14-09-2022 13:27:09	6
1264	7	20220912	12-09-2022 11:18:28	7
1191	8	20221003	03-10-2022 11:34:01	8
5058	9	20220913	13-09-2022 15:41:24	9
4293	10	20220926	26-09-2022 16:44:57	10
1245	11	20221003	03-10-2022 12:02:43	11
7153	12	20221010	10-10-2022 12:40:37	12
6154	13	20220926	26-09-2022 11:29:34	13
1002	14	20221017	17-10-2022 11:36:42	14
1224	15	20221024	24-10-2022 13:08:08	15
1287	16	20221004	04-10-2022 11:49:03	16
5220	17	20221108	08-11-2022 14:53:04	17
6160	18	20221122	22-11-2022 12:47:11	18
7026	19	20221011	11-10-2022 11:40:16	19
8093	20	20221122	22-11-2022 14:52:19	20

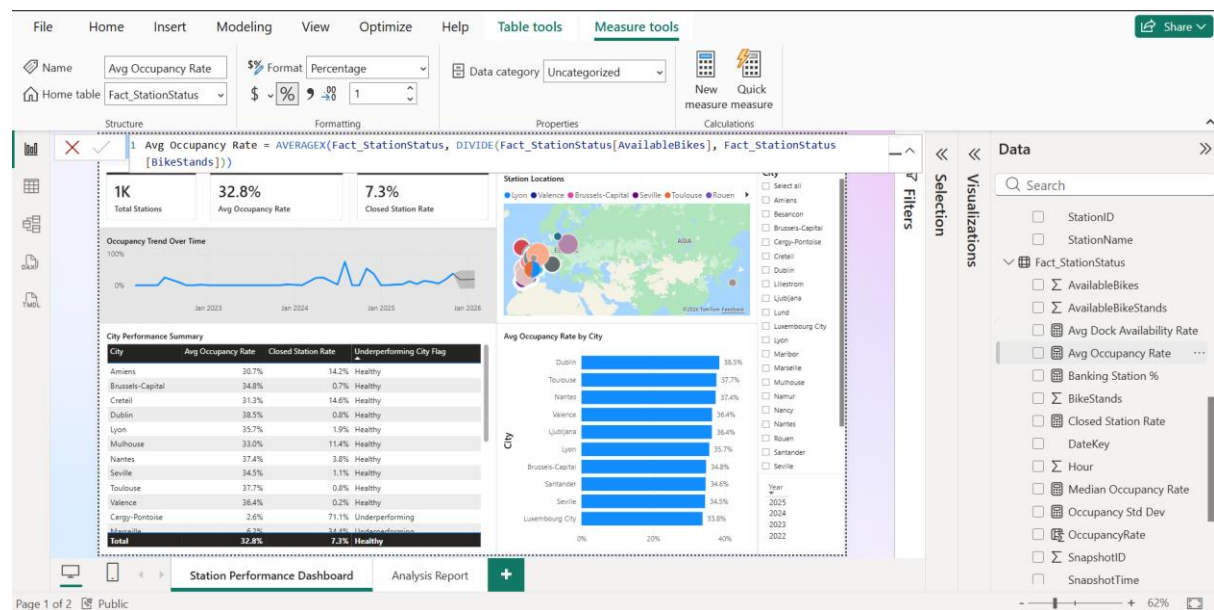
3. Model view - full star schema, all 4 tables, all 3 relationship lines visible



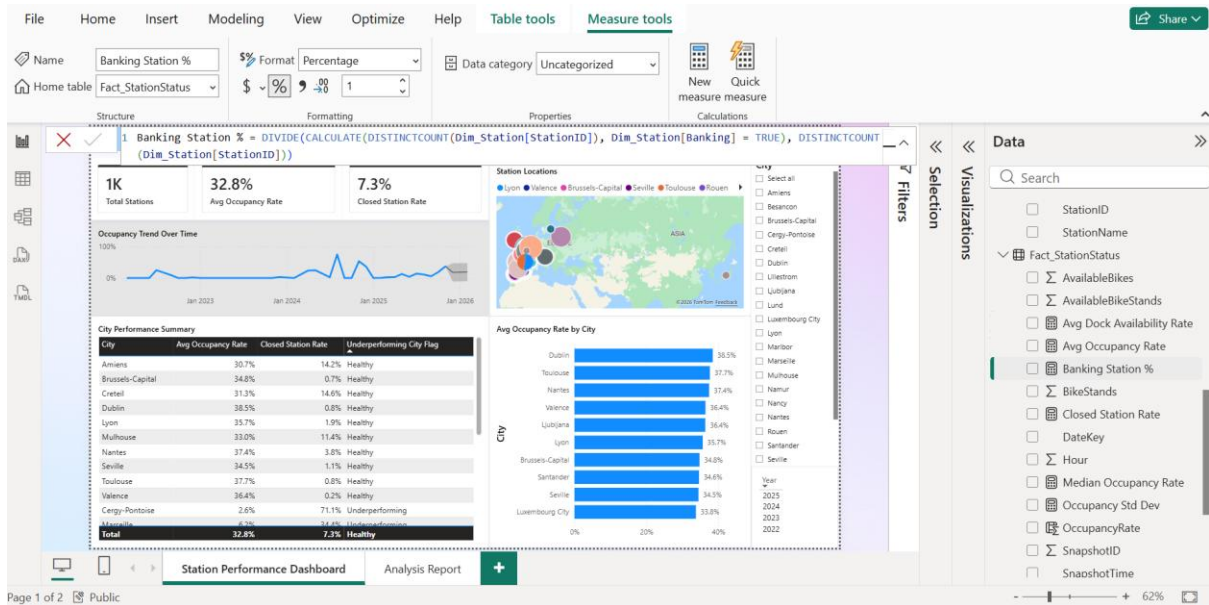
4. DAX:



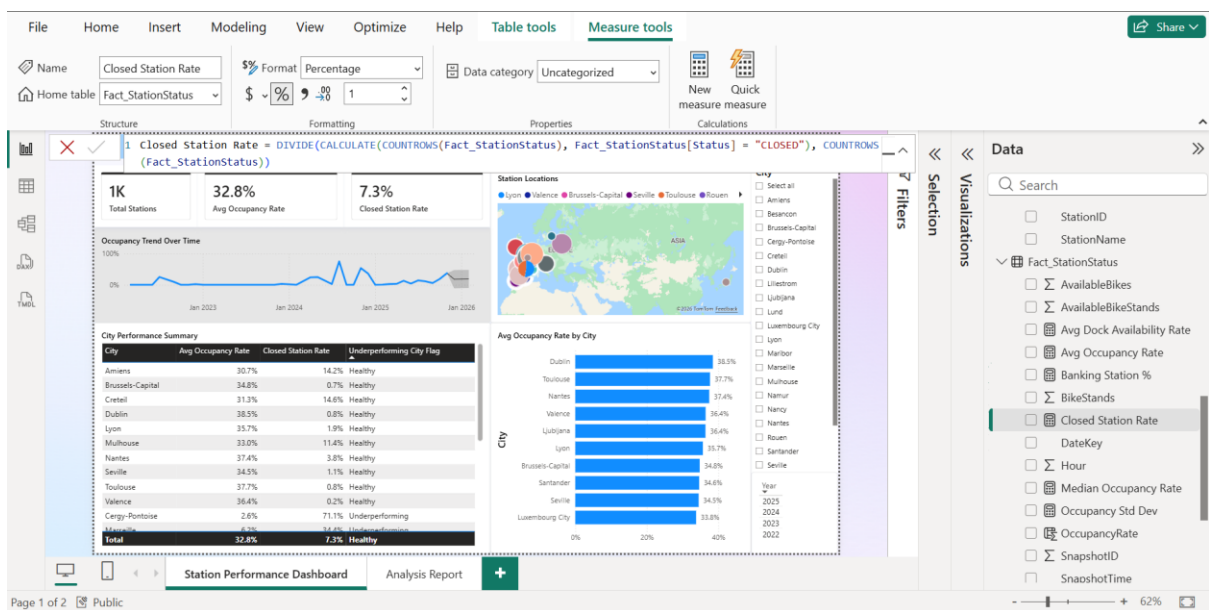
Avg Dock Availability Rate = AVERAGEX(Fact_StationStatus,
DIVIDE(Fact_StationStatus[AvailableBikeStands], Fact_StationStatus[BikeStands]))



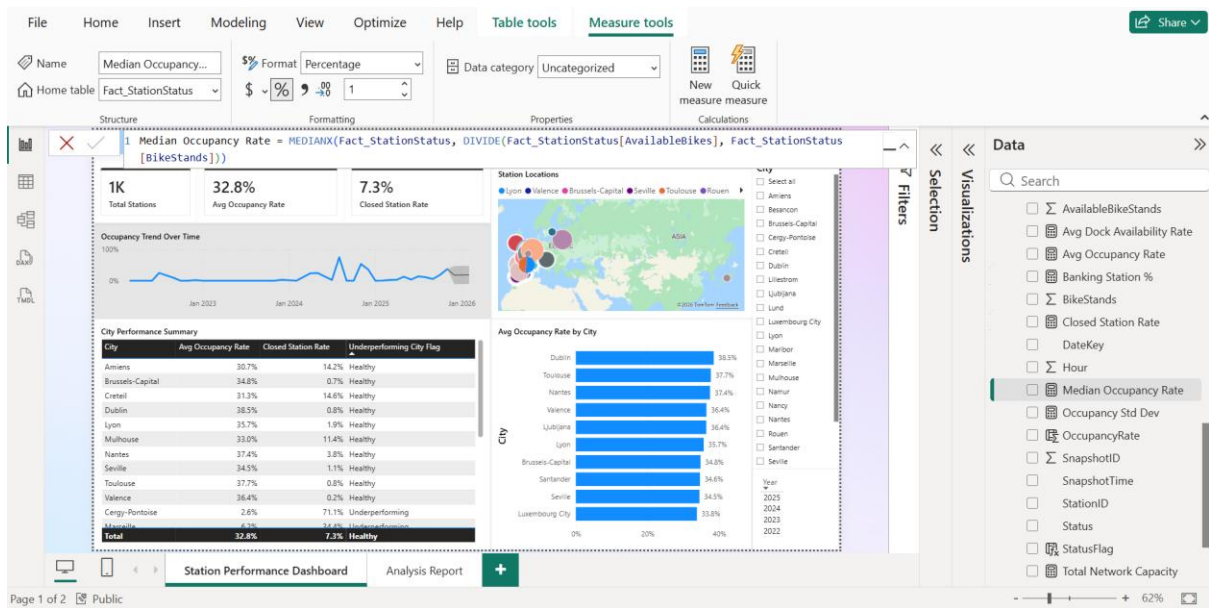
Avg Occupancy Rate = AVERAGEX(Fact_StationStatus,
DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))



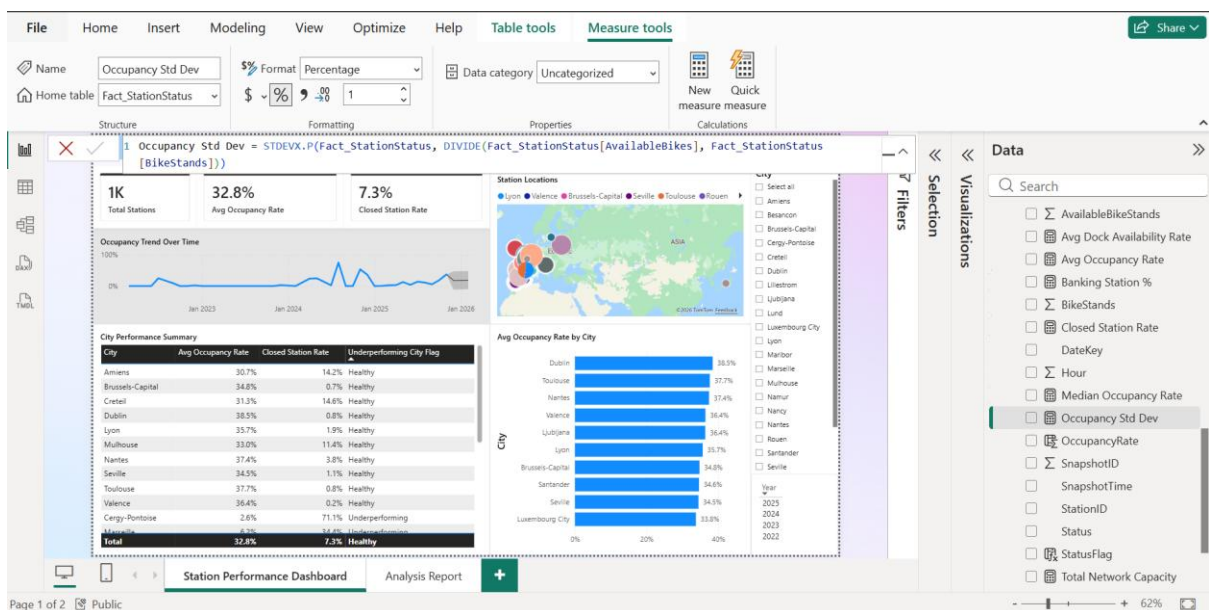
Banking Station % = $\text{DIVIDE}(\text{CALCULATE}(\text{DISTINCTCOUNT}(\text{Dim_Station}[\text{StationID}]), \text{Dim_Station}[\text{Banking}] = \text{TRUE}), \text{DISTINCTCOUNT}(\text{Dim_Station}[\text{StationID}]))$



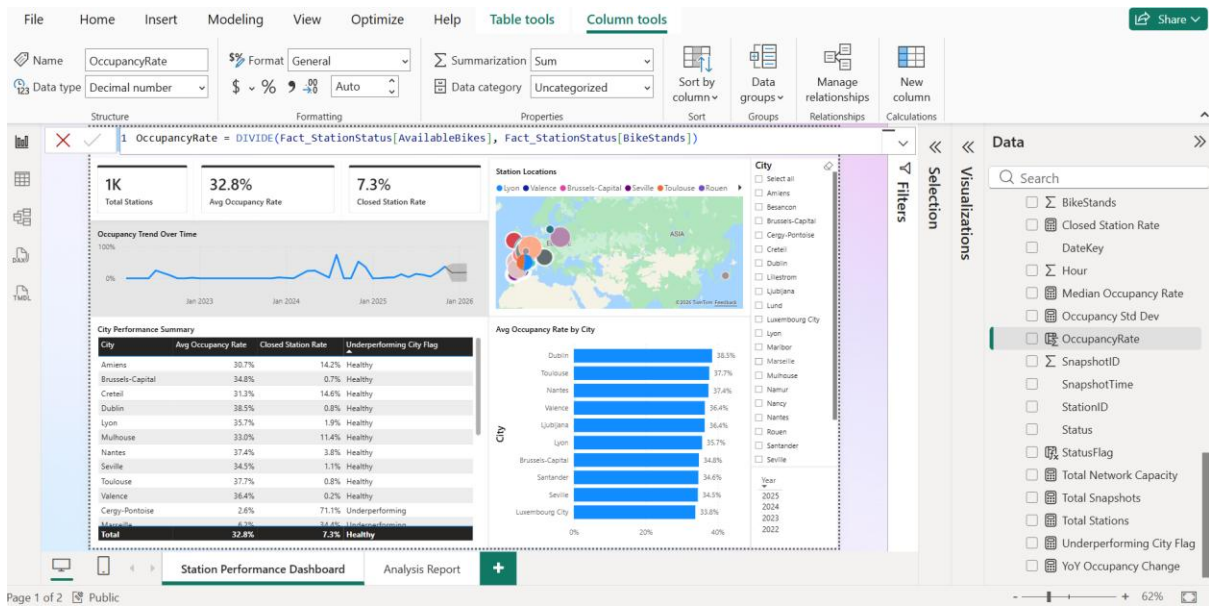
Closed Station Rate = $\text{DIVIDE}(\text{CALCULATE}(\text{COUNTROWS}(\text{Fact_StationStatus}), \text{Fact_StationStatus}[\text{Status}] = \text{"CLOSED"}), \text{COUNTROWS}(\text{Fact_StationStatus}))$



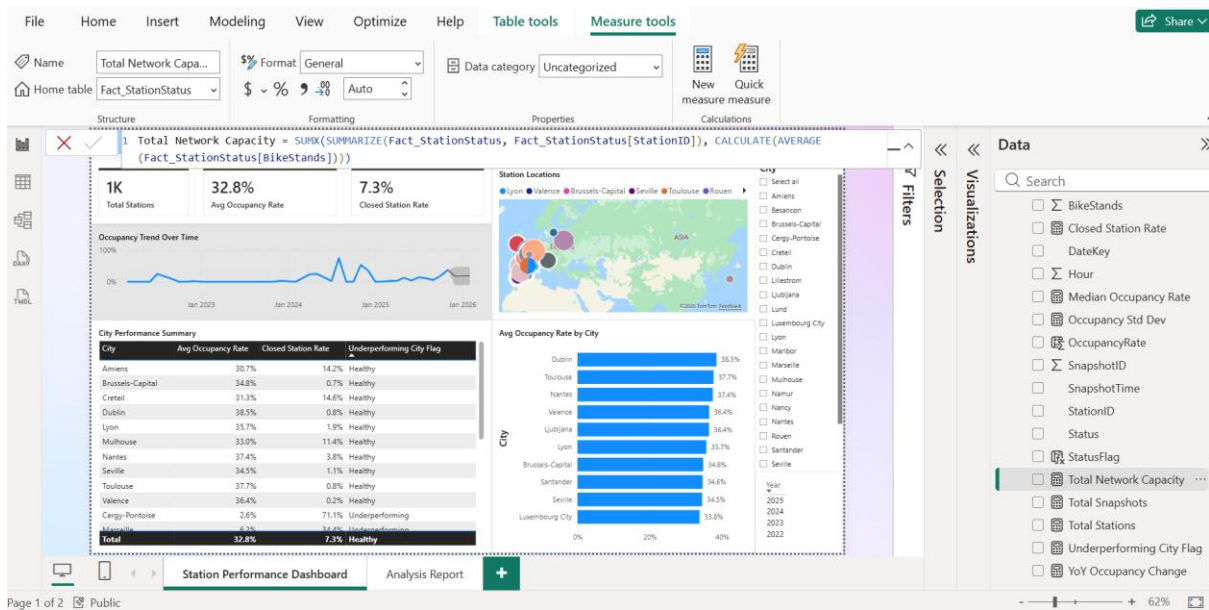
Median Occupancy Rate = MEDIANX(Fact_StationStatus,
DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))



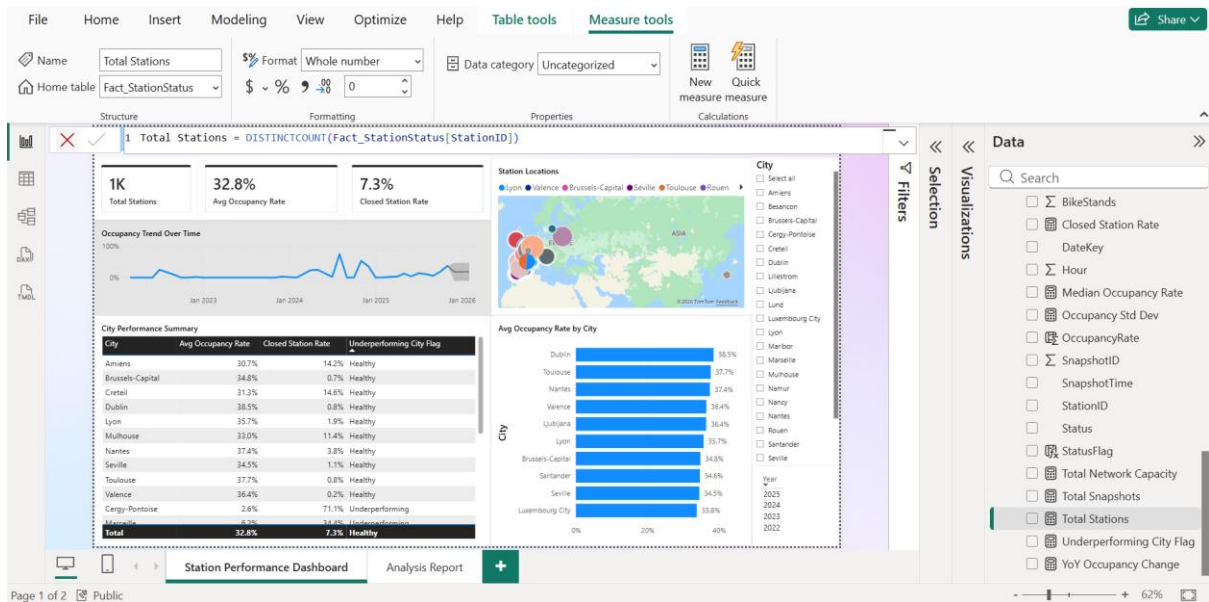
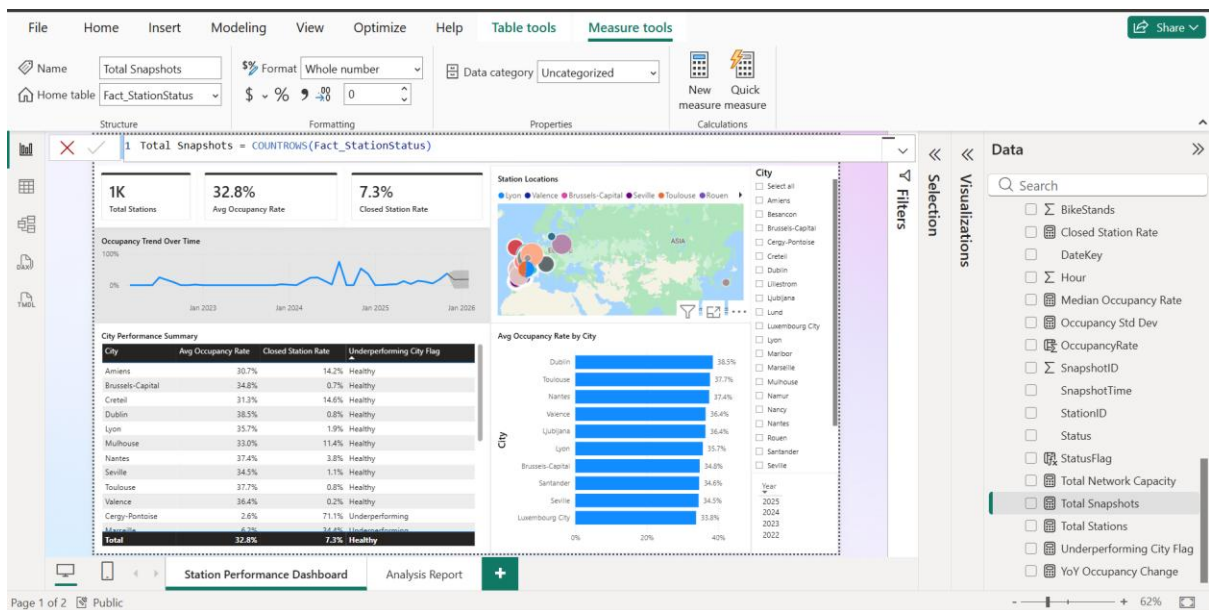
Occupancy Std Dev = STDEVX.P(Fact_StationStatus,
DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands]))

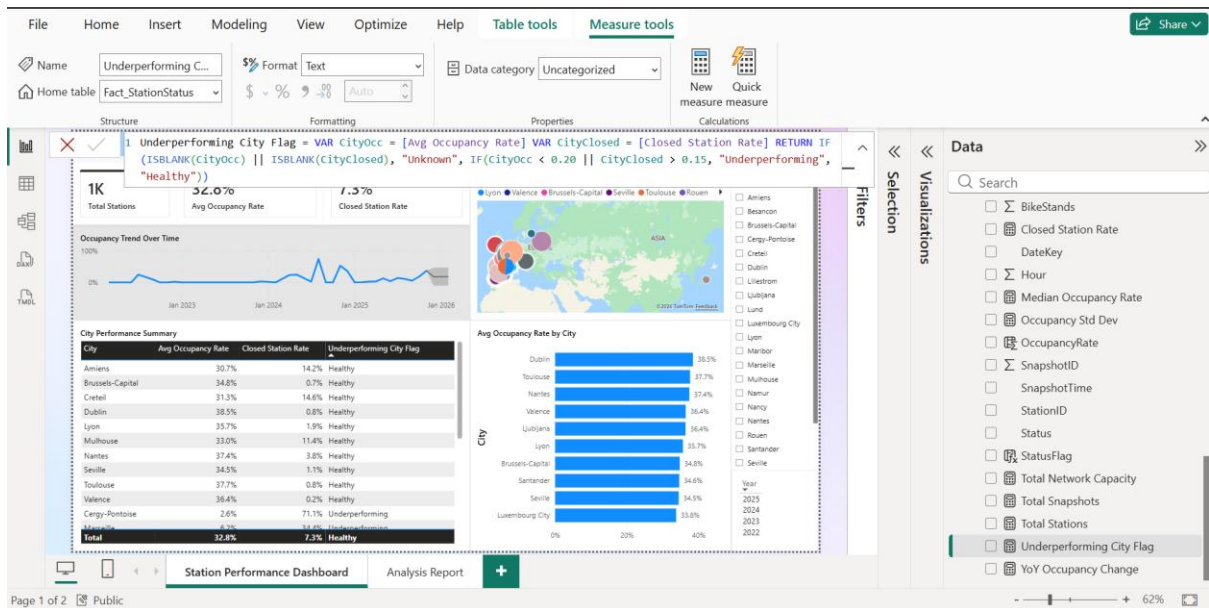


OccupancyRate = DIVIDE(Fact_StationStatus[AvailableBikes], Fact_StationStatus[BikeStands])

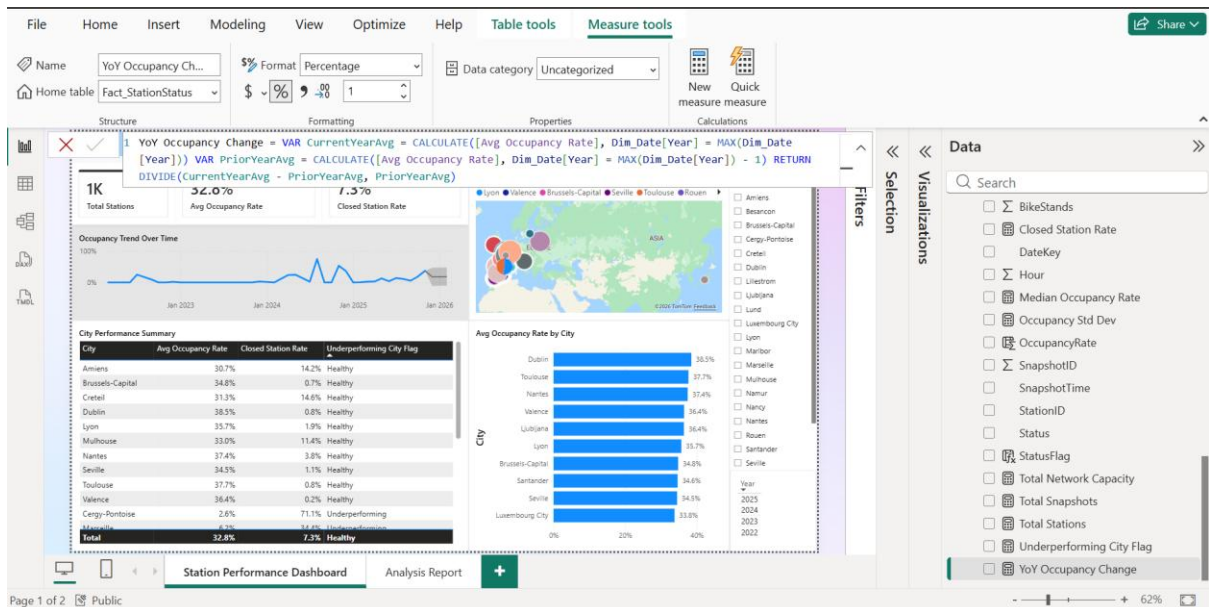


Total Network Capacity = SUMX(SUMMARIZE(Fact_StationStatus, Fact_StationStatus[StationID]), CALCULATE(AVERAGE(Fact_StationStatus[BikeStands])))



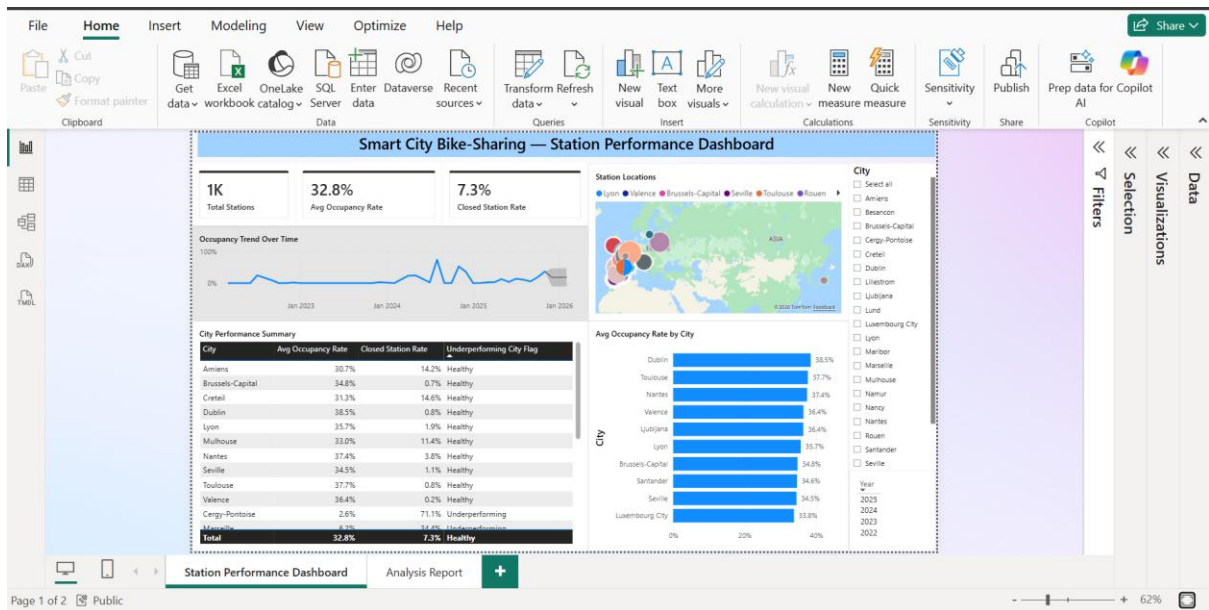


Underperforming City Flag = VAR CityOcc = [Avg Occupancy Rate] VAR CityClosed = [Closed Station Rate] RETURN IF(ISBLANK(CityOcc) || ISBLANK(CityClosed), "Unknown", IF(CityOcc < 0.20 || CityClosed > 0.15, "Underperforming", "Healthy"))



YoY Occupancy Change = VAR CurrentYearAvg = CALCULATE([Avg Occupancy Rate], Dim_Date[Year] = MAX(Dim_Date[Year])) VAR PriorYearAvg = CALCULATE([Avg Occupancy Rate], Dim_Date[Year] = MAX(Dim_Date[Year]) - 1) RETURN DIVIDE(CurrentYearAvg - PriorYearAvg, PriorYearAvg)

Dashboard:



Analysis Report:

